

No.ofFunctionalFeaturesIncludedinthe Solution

Date	3july2026
TeamID	XXXXXX
ProjectName	EduGenie – AI-Powered Educational Assistant
MaximumMarks	5Marks

FunctionalFeaturesOverview

This document presents the functional features implemented in the **EduGenie** project. EduGenie is an AI-powered educational assistant that leverages Generative AI to provide intelligent learning support through question answering, concept explanation, quiz generation, educational text summarization, and personalized learning path recommendations. Each feature has been successfully implemented, tested, and integrated into the overall system, ensuring a smooth and interactive learning experience.

Functional Features Included

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Intelligent Question Answering	Answers educational questions using Generative AI with contextual responses.	Gemini API	Done	5
2	Concept Explanation	Simplifies complex topics into easy-to-understand explanations.	LaMini-Flan-T5	Done	5
3	AI Quiz Generation	Automatically generates topic-based multiple-choice questions.	Gemini API	Done	5
4	Educational Text Summarization	Produces concise summaries of lengthy educational content.	Transformers	Done	5
5	Personalized Learning Path	Creates customized learning roadmaps based on user interests.	Gemini API	Done	5
6	FastAPI Backend	Provides REST APIs to handle all AI	FastAPI	Done	5

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
		functionalities.			
7	Interactive Web Interface	Responsive HTML/CSS frontend for user interaction.	HTML, CSS, Jinja2	Done	5
8	Error Handling & Validation	Handles invalid inputs and API exceptions gracefully.	Python Utilities	Done	5
9	Secure Configuration	Stores API keys and configuration using environment variables.	.env Configuration	Done	5
10	AI Module Integration	Integrates multiple AI models into a unified educational platform.	AI Services	Done	5

Feature Summary

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	10
Core / Must-Have Features	8
Additional Features	2
Features Tested & Verified	10

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Interactive Learning Dashboard
2	Backend / Logic	3	FastAPI APIs, Request Processing, Error Handling
3	AI & NLP Modules	4	Question Answering, Concept Explanation, Quiz Generation, Text Summarization
4	Recommendation System	1	Personalized Learning Path
5	Configuration & Security	1	Environment Variables, API Key Management

